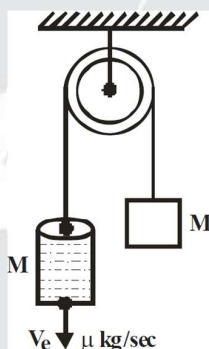
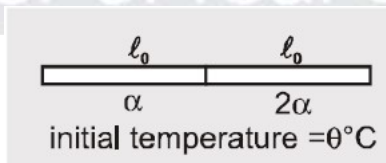


1. A block of mass M and a cylindrical tank which contains water having small hole at bottom, which is closed initially (total mass of cylinder + water is also M), are attached at two ends of an ideal string which passes over an ideal pulley as shown. At $t = 0$ hole is opened such that water starts coming out of the hole with a constant rate μ kg/s and constant velocity V_e relative to the cylinder. Then choose the correct option. (Magnitude of V_e is such that string does not get slack.)

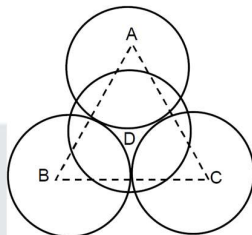


- (A) Magnitude of acceleration of the block at any time t is $\frac{\mu(V_e + gt)}{(2M - \mu t)}$.
- (B) Magnitude of acceleration of the block at any time t is $\frac{\mu(V_e - gt)}{(2M - \mu t)}$.
- (C) Magnitude of acceleration of the block at any time t is $\frac{\mu g t}{2M}$.
- (D) Magnitude of acceleration of the block at any time t is μg
2. Two uniform rods of equal lengths (ℓ_0) and equal masses have coefficient of linear expansion α and 2α are placed in contact on a smooth horizontal surface as shown. The temperature of system is $\theta^\circ\text{C}$. Now the temperature is increased by $\Delta\theta^\circ\text{C}$. The junction of the rods will shift from its initial position by

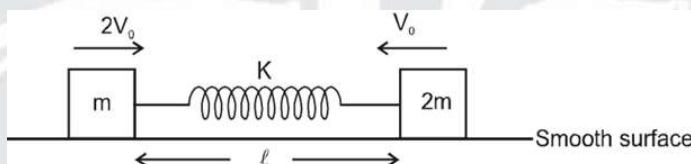


- (A) $\frac{\ell_0 \alpha \Delta\theta}{2}$ (B) $\frac{\ell_0 \alpha \Delta\theta}{3}$ (C) $\frac{\ell_0 \alpha \Delta\theta}{4}$ (D) $\frac{\ell_0 \alpha \Delta\theta}{6}$

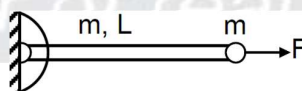
3. Four identical spheres each of mass m and radius R , kept such that 3 spheres are lying in horizontal plane, just in contact with each other and the 4th sphere lies above them, touching rest 3 spheres. Coefficient of friction between each contact surface is same and equal to μ . Top view of the arrangement is shown. The minimum value of μ so that whole system stays in equilibrium will be : (A, B and C are the centers of sphere lying in horizontal plane and D is the centre of 4th sphere which is kept above rest 3 spheres.)



- (A) $\sqrt{3} - \sqrt{2}$ (B) $\frac{\sqrt{3}-1}{2}$ (C) $\sqrt{2} - 1$ (D) $\frac{\sqrt{2}-1}{2}$
4. A sphere and a cube of same material and same total surface area are placed in same evacuated space after they are heated to same temperature. The ratio of their initial rate of cooling in space is :
- (A) 1 (B) $\sqrt{\pi}$ (C) $\sqrt{\frac{\pi}{6}}$ (D) $\frac{1}{\sqrt{6}}$
5. Two blocks A & B of masses m and $2m$ respectively, attached at opposite ends of a spring of spring constant K , placed on smooth horizontal surface. Spring is initially at its natural length ℓ . A is given a velocity $2V_0$ and B given velocity V_0 as shown.

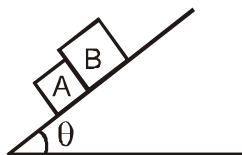


- Maximum separation between m and centre of mass of the system will be :
- (A) $\frac{\ell}{3} + \sqrt{\frac{8mV_0^2}{3K}}$ (B) $\frac{\ell}{3} + \sqrt{\frac{2mV_0^2}{3K}}$ (C) $\frac{2\ell}{3} + \sqrt{\frac{2mV_0^2}{3K}}$ (D) $\frac{2\ell}{3} + \sqrt{\frac{3mV_0^2}{3K}}$
6. A thin uniform rod of mass m and length L is hinged at one end and is free to rotate in the vertical plane. A small ball of mass m is fixed to the other end of the rod. A constant horizontal force $F = mg$ is also acting as shown in figure.

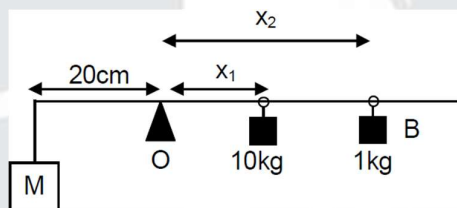


- (A) Acceleration of the ball at the instant of releasing the rod is $9g/8$.
- (B) Hinge force of $\frac{5mg}{16}$ acts in vertical direction just after releasing rod.
- (C) Speed of the ball when the rod becomes vertical is $\sqrt{\frac{3g\ell}{2}}$.
- (D) Speed of the ball when the rod becomes vertical is $\sqrt{\frac{3g\ell}{4}}$.

7. The two blocks A and B of equal mass are initially in contact when released from rest on the inclined plane. The coefficients of friction between the inclined plane and A is μ_1 and that between inclined plane and B is μ_2 . (Assume $\tan \theta > \mu_1$ and μ_2)

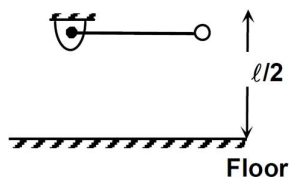


- (A) If $\mu_1 > \mu_2$, the blocks will always remain in contact.
- (B) If $\mu_1 < \mu_2$, the blocks will slide down with different accelerations
- (C) If $\mu_1 > \mu_2$, the blocks will have a common acceleration $\frac{1}{2}(\mu_1 + \mu_2) g \sin \theta$
- (D) If $\mu_1 < \mu_2$, the blocks will have a common acceleration $\frac{\mu_1 \mu_2}{\mu_1 + \mu_2} g \sin \theta$
8. A beam balance with light rod is used to determine the unknown mass M. The rod is hinged from O. Two standard masses A (10kg) and B(1kg) which can slide on the rod are set at distances x_1 and x_2 from O to balance the rod in horizontal position and they are measured with maximum permissible error 0.1cm.

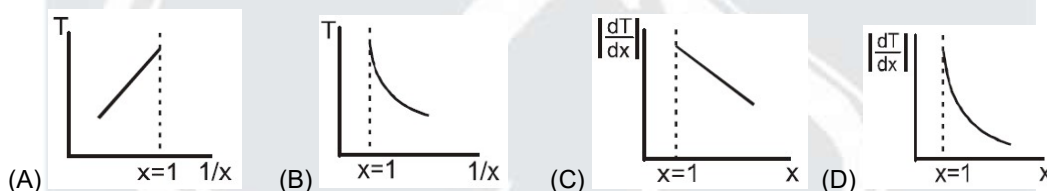
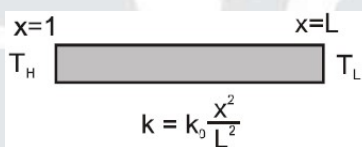


- Choose the correct option(s) :
- Select correct alternatives
- (A) The maximum permissible error in M depends on absolute values of x_1 and x_2 .
- (B) If $x_1 = 5.0$ cm and $x_2 = 10.0$ cm then the maximum permissible error in M is 0.055kg
- (C) If $x_1 = 10.0$ cm and $x_2 = 20.0$ cm then percentage error in measurement of M is approximately 0.9%
- (D) The percentage error in measurement of M depends on absolute values of x_1 and x_2
9. Suppose a body that is acted on by only two forces, is accelerated. For this situation, mark the incorrect statement(s) :
- (A) The body cannot move with constant speed
- (B) The velocity can never be zero
- (C) The sum of two forces cannot be zero
- (D) The two forces must act in the same line

10. A bob of mass m connected to the end of an inextensible string of length ℓ , is released from position shown in figure. If impacts of bob with smooth floor is perfectly inelastic. Choose the correct option(s).

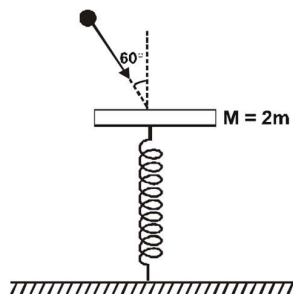


- (A) The maximum height reached by the bob during subsequent motion is $\frac{\ell}{32}$.
- (B) Maximum height reached by the bob during subsequent motion is zero.
- (C) Energy loss from the time duration bob is released to the time it reaches at its maximum height during subsequent motion is $\frac{3mg\ell}{8}$.
- (D) Energy loss from the time duration bob is released to the time it reaches at its maximum height during subsequent motion is $\frac{15mg\ell}{32}$.
11. A rod of uniform cross-section but non-uniform thermal conductivity which vary as $k = k_0 \frac{x^2}{L^2}$ ($1 \leq x \leq L$) (as shown in figure) is kept between fixed temperature difference for a long time. Select the correct option(s) :

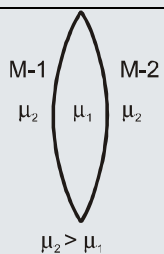
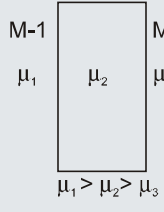
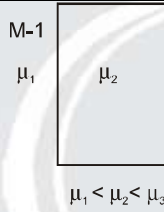
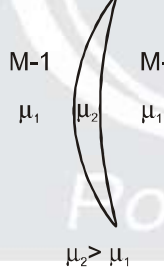


COMPREHENSION

A particle of mass m collides elastically with the pan of mass ($M = 2m$) of a spring balance, as shown in figure. Pan is in equilibrium before collision. Spring constant is k and speed of the particle before collision is v_0 . Answer the following three questions regarding this collision.



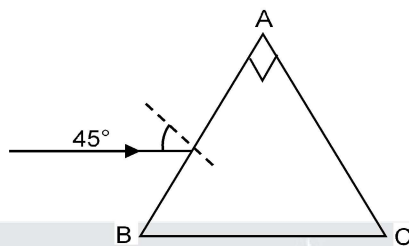
12. Maximum height attained by the particle from the point of collision after collision is
 (A) $\frac{v_0^2}{16g}$ (B) $\frac{v_0^2}{8g}$ (C) $\frac{v_0^2}{36g}$ (D) $\frac{v_0^2}{72g}$
13. Minimum kinetic energy of the particle after collision is
 (A) $\frac{mv_0^2}{8}$ (B) $\frac{3mv_0^2}{8}$ (C) $\frac{3mv_0^2}{4}$ (D) $\frac{mv_0^2}{2}$
14. In List-I an optical system is shown. List-II tells about optical events. Match the proper entries from List-II to List-I and select the correct answer using the code given below.

List-I		List-II	
(P)		(1)	Parallel beam of light which incidents from medium M-1 emerges out as converging in medium M-2
(Q)		(2)	Parallel beam of light which incidents from medium M-1 at angle $i \neq 0$ emerges out as parallel beam in medium M-2, will have lesser diameter than the diameter of incident beam. (Assume No TIR)
(R)		(3)	Parallel beam of light which incidents from medium M-1 emerges out as diverging in medium M-2.
(S)		(4)	Parallel beam of light which incidents from medium M-1 at angle $i \neq 0$ emerges out as parallel beam in medium M-2 will have larger diameter than the diameter of incident beam. (Assume No TIR)

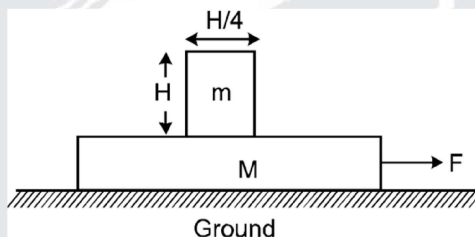
Code :

	P	Q	R	S
(A)	3	4	2	1
(B)	3	2	4	1
(C)	2	4	1	3
(D)	3	4	1	2

15. Monochromatic light falls on a right angled prism at an angle of 45° and the emergent ray is found to move along AC tangentially. The refraction index of material of prism is $\sqrt{\frac{x}{y}}$ find $x + y = ?$



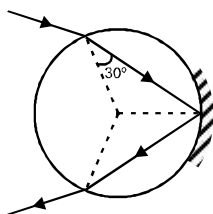
16. A block of mass $m = 2\text{kg}$ of shown dimensions is placed on a plank of mass $M = 6\text{Kg}$ which is placed on smooth horizontal plane. The coefficient of friction between the block and the plank is $\mu = \frac{1}{3}$. If a horizontal force F is applied on the plank, then find the maximum value of F (in N) for which the block and the plank move together.



17. A particle is projected up an incline plane of inclination β at an elevation of α to the horizontal. Find the ratio of $\tan \alpha$ and $\tan \beta$, if the particle strikes the plane Horizontally
18. A man does work W on a ball to throw it vertically upto a height h on earth, if man does work $\frac{9W}{k}$ on same ball to throw it vertically upto height $2h$ on another planet having its radius $\frac{1}{4}$ of radius of earth and planet has same density as that of earth. Find 'k' ? (h is very small in comparison to radius of earth).

19. Two particles A & B are located at points $(0, -20\sqrt{3})$ and $(0,0)$ in $x - y$ plane. They start moving simultaneously at $t = 0$, with constant velocities $\vec{v}_A = 5 \hat{i} \text{ m/s}$ and $\vec{v}_B = -5\sqrt{3} \hat{j} \text{ m/s}$ respectively. Time when they are closest to each other is (in sec)

20. A ray is incident on a sphere of refractive index $\sqrt{2}$ as shown in figure. Angle of refraction of the ray inside sphere is 30° . If total deviation suffered by the ray is $30x^\circ$, then find x.



ANSWER KEY

1. (A)	2. (C)	3. (A)	4. (C)	5. (D)
6. (ABD)	7. (AB)	8. (BCD)	9. (ABD)	10. (AD)
11. (AD)	12. (D)	13. (D)	14. (A)	15. 5
16. $f_{\max} = (M+m) \frac{g}{4}$	17. 2	18. 18.00	19. 3	20. 5